Linear Algebra and Its Applications

Section 5.5: Complex eigenvalues

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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5.5 Complex eigenvalues

A real matrix can have nonreal eigenvalues when it acts on complex coordinate vectors. Complex arithmetic uses $i^2 = -1$ and conjugation $\overline{a+bi} = a-bi$. For a real matrix, conjugating $Av = \lambda v$ gives $A\overline{v} = \overline{\lambda}\,\overline{v}$, so nonreal eigenvalues occur in conjugate pairs.

The real matrix $C = \left[\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right]$ is a scaling followed by a rotation: $C = r\left[\begin{smallmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{smallmatrix}\right]$, where $r = \sqrt{a^2+b^2}$, $\cos\varphi = a/r$, and $\sin\varphi = b/r$. The sign of b matters for clockwise versus counterclockwise rotation. We use angles in $(-\pi, \pi]$.

Practice problem 5.5.P1: Verify the two complex eigenpairs directly

Multiplication gives

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix}\begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} a+bi \\ b-ai \end{bmatrix} = (a+bi)\begin{bmatrix} 1 \\ -i \end{bmatrix}.$$

Likewise $A(1,i)^T = (a-bi)(1,i)^T$. Both vectors are nonzero, so these are genuine eigenpairs. They are conjugate to one another. If $b = 0$, the eigenvalues coincide at a; the verified equations still hold, and in fact every nonzero vector is an eigenvector of aI_2.

Exercise 5.5.1: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}.$$

The characteristic equation is $t^2 - 4t + 5 = 0$, whose two roots are $2-i,\ 2+i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 2-i$, the first equation of $(A-\lambda I)x = 0$ is

$$(-1+i)x_1 + (-2)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -2s$ and $x_2 = (1-i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{2-i} = \mathrm{Span}_{\mathbb{C}}\left\{\begin{bmatrix} -2 \\ 1-i \end{bmatrix}\right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 2+i$, the first equation of $(A-\lambda I)x = 0$ is

$$(-1-i)x_1 + (-2)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -2s$ and $x_2 = (1+i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{2+i} = \mathrm{Span}_{\mathbb{C}}\left\{\begin{bmatrix} -2 \\ 1+i \end{bmatrix}\right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace.

Exercise 5.5.2: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 3 & -3 \\ 3 & 3 \end{bmatrix}.$$

The characteristic equation is $t^2 - 6t + 18 = 0$, whose two roots are $3 - 3i,\ 3 + 3i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 3 - 3i$, the first equation of $(A - \lambda I)x = 0$ is

$$(3i)x_1 + (-3)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -3s$ and $x_2 = (-3i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{3-3i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -3 \\ -3i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 3 + 3i$, the first equation of $(A - \lambda I)x = 0$ is

$$(-3i)x_1 + (-3)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -3s$ and $x_2 = (3i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{3+3i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -3 \\ 3i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace.

Exercise 5.5.3: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 5 & 1 \\ -8 & 1 \end{bmatrix}.$$

The characteristic equation is $t^2 - 6t + 13 = 0$, whose two roots are $3 - 2i,\ 3 + 2i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 3 - 2i$, the first equation of $(A - \lambda I)x = 0$ is

$$(2 + 2i)x_1 + (1)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = 1s$ and $x_2 = (-2 - 2i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{3-2i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} 1 \\ -2 - 2i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 3 + 2i$, the first equation of $(A - \lambda I)x = 0$ is

$$(2 - 2i)x_1 + (1)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = 1s$ and $x_2 = (-2 + 2i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{3+2i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} 1 \\ -2 + 2i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace.

Exercise 5.5.4: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}.$$

The characteristic equation is $t^2 - 4t + 5 = 0$, whose two roots are $2 - i,\ 2 + i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 2 - i$, the first equation of $(A - \lambda I)x = 0$ is

$$(-1 + i)x_1 + (-2)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -2s$ and $x_2 = (1 - i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{2-i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -2 \\ 1 - i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 2 + i$, the first equation of $(A - \lambda I)x = 0$ is

$$(-1 - i)x_1 + (-2)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -2s$ and $x_2 = (1 + i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{2+i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -2 \\ 1 + i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. The printed matrix in Exercise 4 is identical to that in Exercise 1; the repeated answer reflects the actual supplied edition, rather than an altered matrix.

Exercise 5.5.5: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 5 \end{bmatrix}.$$

The characteristic equation is $t^2 - 8t + 17 = 0$, whose two roots are $4 - i,\ 4 + i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 4 - i$, the first equation of $(A - \lambda I)x = 0$ is

$$(-1 + i)x_1 + (1)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = 1s$ and $x_2 = (1 - i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{4-i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} 1 \\ 1 - i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 4 + i$, the first equation of $(A - \lambda I)x = 0$ is

$$(-1 - i)x_1 + (1)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = 1s$ and $x_2 = (1 + i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{4+i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} 1 \\ 1 + i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace.

Exercise 5.5.6: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 7 & -5 \\ 1 & 3 \end{bmatrix}.$$

The characteristic equation is $t^2 - 10t + 26 = 0$, whose two roots are $5 - i,\ 5 + i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 5 - i$, the first equation of $(A - \lambda I)x = 0$ is

$$(2 + i)x_1 + (-5)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -5s$ and $x_2 = (-2 - i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{5-i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -5 \\ -2 - i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 5 + i$, the first equation of $(A - \lambda I)x = 0$ is

$$(2 - i)x_1 + (-5)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -5s$ and $x_2 = (-2 + i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{5+i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -5 \\ -2 + i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace.

Exercise 5.5.7: Rotation angle and positive scale

The matrix has the canonical entries $a = \sqrt{3}$ and $b = 1$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = \sqrt{3} \pm (1)i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = 2.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = \frac{\sqrt{3}}{2}$ and $\sin\varphi = b/r = \frac{1}{2}$. In the requested interval these give

$$\varphi = \frac{\pi}{6}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.8: Rotation angle and positive scale

The matrix has the canonical entries $a = 3$ and $b = -3\sqrt{3}$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = 3 \pm (-3\sqrt{3})i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = 6.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = \frac{1}{2}$ and $\sin\varphi = b/r = -\frac{\sqrt{3}}{2}$. In the requested interval these give

$$\varphi = -\frac{\pi}{3}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.9: Rotation angle and positive scale

The matrix has the canonical entries $a = 0$ and $b = -2$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = 0 \pm (-2)i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = 2.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = 0$ and $\sin\varphi = b/r = -1$. In the requested interval these give

$$\varphi = -\frac{\pi}{2}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.10: Rotation angle and positive scale

The matrix has the canonical entries $a = 0$ and $b = -\frac{1}{2}$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = 0 \pm (-\frac{1}{2})i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = \frac{1}{2}.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = 0$ and $\sin\varphi = b/r = -1$. In the requested interval these give

$$\varphi = -\frac{\pi}{2}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.11: Rotation angle and positive scale

The matrix has the canonical entries $a = -\sqrt{3}$ and $b = -1$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = -\sqrt{3} \pm (-1)i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = 2.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = -\frac{\sqrt{3}}{2}$ and $\sin\varphi = b/r = -\frac{1}{2}$. In the requested interval these give

$$\varphi = -\frac{5\pi}{6}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.12: Rotation angle and positive scale

The matrix has the canonical entries $a = 3$ and $b = \sqrt{3}$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = 3 \pm (\sqrt{3})i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = 2\sqrt{3}.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = \frac{\sqrt{3}}{2}$ and $\sin\varphi = b/r = \frac{1}{2}$. In the requested interval these give

$$\varphi = \frac{\pi}{6}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.13: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}.$$

The characteristic roots are $2 \pm (1)i$. Choose the root $\lambda = a - bi = 2 - i$ with $b = 1 > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -2 \\ 1 - i \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Equate real and imaginary parts in $A(u + iw) = (a - bi)(u + iw)$. This yields $Au = au + bw$ and $Aw = -bu + aw$, so in the ordered real basis (u, w) the matrix is C. Consequently take

$$P = \begin{bmatrix} -2 & 0 \\ 1 & -1 \end{bmatrix}, \qquad C = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}.$$

Here $\det P = 2 \neq 0$. Direct exact multiplication verifies $AP = PC$, which, after multiplying by P^{-1}, gives the required $A = PCP^{-1}$.

Exercise 5.5.14: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} 3 & -3 \\ 1 & 1 \end{bmatrix}.$$

The characteristic roots are $2 \pm (\sqrt{2})i$. Choose the root $\lambda = a - bi = 2 - \sqrt{2}i$ with $b = \sqrt{2} > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -3 \\ -1 - \sqrt{2}i \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -3 \\ -1 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -\sqrt{2} \end{bmatrix}.$$

Equate real and imaginary parts in $A(u+iw)=(a-bi)(u+iw)$. This yields $Au=au+bw$ and $Aw=-bu+aw$, so in the ordered real basis (u,w) the matrix is C. Consequently take

$$P=\begin{bmatrix}-3 & 0\\ -1 & -\sqrt{2}\end{bmatrix},\qquad C=\begin{bmatrix}2 & -\sqrt{2}\\ \sqrt{2} & 2\end{bmatrix}.$$

Here $\det P=3\sqrt{2}\neq 0$. Direct exact multiplication verifies $AP=PC$, which, after multiplying by P^{-1}, gives the required $A=PCP^{-1}$.

Exercise 5.5.15: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A=\begin{bmatrix}0 & 5\\ -2 & 2\end{bmatrix}.$$

The characteristic roots are $1\pm(3)i$. Choose the root $\lambda=a-bi=1-3i$ with $b=3>0$. Solving its first homogeneous equation as above gives the eigenvector

$$v=\begin{bmatrix}5\\ 1-3i\end{bmatrix}=u+iw,\qquad u=\begin{bmatrix}5\\ 1\end{bmatrix},\quad w=\begin{bmatrix}0\\ -3\end{bmatrix}.$$

Equate real and imaginary parts in $A(u+iw)=(a-bi)(u+iw)$. This yields $Au=au+bw$ and $Aw=-bu+aw$, so in the ordered real basis (u,w) the matrix is C. Consequently take

$$P=\begin{bmatrix}5 & 0\\ 1 & -3\end{bmatrix},\qquad C=\begin{bmatrix}1 & -3\\ 3 & 1\end{bmatrix}.$$

Here $\det P=-15\neq 0$. Direct exact multiplication verifies $AP=PC$, which, after multiplying by P^{-1}, gives the required $A=PCP^{-1}$.

Exercise 5.5.16: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A=\begin{bmatrix}4 & -2\\ 1 & 6\end{bmatrix}.$$

The characteristic roots are $5\pm(1)i$. Choose the root $\lambda=a-bi=5-i$ with $b=1>0$. Solving its first homogeneous equation as above gives the eigenvector

$$v=\begin{bmatrix}-2\\ 1-i\end{bmatrix}=u+iw,\qquad u=\begin{bmatrix}-2\\ 1\end{bmatrix},\quad w=\begin{bmatrix}0\\ -1\end{bmatrix}.$$

Equate real and imaginary parts in $A(u+iw)=(a-bi)(u+iw)$. This yields $Au=au+bw$ and $Aw=-bu+aw$, so in the ordered real basis (u,w) the matrix is C. Consequently take

$$P=\begin{bmatrix}-2 & 0\\ 1 & -1\end{bmatrix},\qquad C=\begin{bmatrix}5 & -1\\ 1 & 5\end{bmatrix}.$$

Here $\det P=2\neq 0$. Direct exact multiplication verifies $AP=PC$, which, after multiplying by P^{-1}, gives the required $A=PCP^{-1}$.

Exercise 5.5.17: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} -11 & -4 \\ 20 & 5 \end{bmatrix}.$$

The characteristic roots are $-3 \pm (4)i$. Choose the root $\lambda = a - bi = -3 - 4i$ with $b = 4 > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -4 \\ 8 - 4i \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -4 \\ 8 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -4 \end{bmatrix}.$$

Equate real and imaginary parts in $A(u + iw) = (a - bi)(u + iw)$. This yields $Au = au + bw$ and $Aw = -bu + aw$, so in the ordered real basis (u, w) the matrix is C. Consequently take

$$P = \begin{bmatrix} -4 & 0 \\ 8 & -4 \end{bmatrix}, \qquad C = \begin{bmatrix} -3 & -4 \\ 4 & -3 \end{bmatrix}.$$

Here $\det P = 16 \neq 0$. Direct exact multiplication verifies $AP = PC$, which, after multiplying by P^{-1}, gives the required $A = PCP^{-1}$.

Exercise 5.5.18: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} 3 & -5 \\ 2 & 5 \end{bmatrix}.$$

The characteristic roots are $4 \pm (3)i$. Choose the root $\lambda = a - bi = 4 - 3i$ with $b = 3 > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -5 \\ 1 - 3i \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -5 \\ 1 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -3 \end{bmatrix}.$$

Equate real and imaginary parts in $A(u + iw) = (a - bi)(u + iw)$. This yields $Au = au + bw$ and $Aw = -bu + aw$, so in the ordered real basis (u, w) the matrix is C. Consequently take

$$P = \begin{bmatrix} -5 & 0 \\ 1 & -3 \end{bmatrix}, \qquad C = \begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}.$$

Here $\det P = 15 \neq 0$. Direct exact multiplication verifies $AP = PC$, which, after multiplying by P^{-1}, gives the required $A = PCP^{-1}$.

Exercise 5.5.19: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} \frac{38}{25} & -\frac{7}{10} \\ \frac{14}{25} & \frac{2}{5} \end{bmatrix}.$$

The characteristic roots are $\frac{24}{25} \pm (\frac{7}{25})i$. Choose the root $\lambda = a - bi = \frac{24}{25} - \frac{7i}{25}$ with $b = \frac{7}{25} > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -\frac{7}{10} \\ -\frac{14}{25} - \frac{7i}{25} \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -\frac{7}{10} \\ -\frac{14}{25} \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -\frac{7}{25} \end{bmatrix}.$$

Equate real and imaginary parts in $A(u + iw) = (a - bi)(u + iw)$. This yields $Au = au + bw$ and $Aw = -bu + aw$, so in the ordered real basis (u, w) the matrix is C. Consequently take

$$P = \begin{bmatrix} -\frac{7}{10} & 0 \\ -\frac{14}{25} & -\frac{7}{25} \end{bmatrix}, \qquad C = \begin{bmatrix} \frac{24}{25} & -\frac{7}{25} \\ \frac{7}{25} & \frac{24}{25} \end{bmatrix}.$$

Here $\det P = \frac{49}{250} \neq 0$. Direct exact multiplication verifies $AP = PC$, which, after multiplying by P^{-1}, gives the required $A = PCP^{-1}$. All decimals in the question were treated as exact terminating rationals, so these entries are exact, not rounded numerical eigenpairs.

Exercise 5.5.20: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} -3 & -8 \\ 4 & 5 \end{bmatrix}.$$

The characteristic roots are $1 \pm (4)i$. Choose the root $\lambda = a - bi = 1 - 4i$ with $b = 4 > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -8 \\ 4 - 4i \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -8 \\ 4 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -4 \end{bmatrix}.$$

Equate real and imaginary parts in $A(u + iw) = (a - bi)(u + iw)$. This yields $Au = au + bw$ and $Aw = -bu + aw$, so in the ordered real basis (u, w) the matrix is C. Consequently take

$$P = \begin{bmatrix} -8 & 0 \\ 4 & -4 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & -4 \\ 4 & 1 \end{bmatrix}.$$

Here $\det P = 32 \neq 0$. Direct exact multiplication verifies $AP = PC$, which, after multiplying by P^{-1}, gives the required $A = PCP^{-1}$.

Exercise 5.5.21: Conjugation and real invariant directions

The first equation in Example 2 is $(-0.3 + 0.6i)x_1 - 0.6x_2 = 0$. Solve it for the second component:

$$x_2 = \frac{-0.3 + 0.6i}{0.6} x_1 = (-1/2 + i)x_1.$$

Taking $x_1 = 2$ gives $y = (2, -1 + 2i)^T$, as claimed. The original vector was $v_1 = (-2 - 4i, 5)^T$. With $c = (-1 + 2i)/5$, direct multiplication gives

$$c(-2 - 4i) = 2, \qquad c \cdot 5 = -1 + 2i.$$

Thus $y = cv_1$, with $c \neq 0$. They span the same complex eigenspace; different choices of a free parameter do not produce different eigenspace directions.

Exercise 5.5.22: Conjugation and real invariant directions

Suppose $Ax = \lambda x$ with $x \neq 0$ and choose a nonzero complex scalar μ. Complex linearity gives $A(\mu x) = \mu Ax = \mu\lambda x = \lambda(\mu x)$. Also $\mu x \neq 0$, because multiplying a zero result by $1/\mu$ would imply $x = 0$. Thus μx is an eigenvector for the same eigenvalue. Both its eigenvector equation and nonzero status are required.

Exercise 5.5.23: Conjugation and real invariant directions

Write $x^* = \overline{x}^T$ and $q = x^*Ax$, a 1×1 scalar. Here is the complete chain with the reasons for all five transitions:

$$\begin{aligned}
\overline{q} &= \overline{\overline{x}^T A x} \\
&= x^T \overline{A}\,\overline{x} && \text{(a): conjugation distributes over products;} \\
&= x^T A \overline{x} && \text{(b): } A \text{ has real entries;} \\
&= (x^T A \overline{x})^T && \text{(c): a scalar equals its transpose;} \\
&= \overline{x}^T A^T x && \text{(d): transpose reverses product order;} \\
&= \overline{x}^T A x = q && \text{(e): } A^T = A.
\end{aligned}$$

A complex number equals its conjugate exactly when its imaginary part is zero, so q is real. Conjugation does not reverse product order in (a); transposition does in (d).

Exercise 5.5.24: Conjugation and real invariant directions

Prove the eigenvalue is real without assuming a spectral theorem.

Let $x \neq 0$ and $Ax = \lambda x$. Multiply on the left by $x^* = \overline{x}^T$:

$$x^*Ax = \lambda x^*x, \qquad x^*x = \sum_{j=1}^{n} |x_j|^2 > 0.$$

The numerator is real by Exercise 23, so $\lambda = (x^*Ax)/(x^*x)$ is a real number.

Separate the real and imaginary components, checking nonzero status.

Write $x = u + iv$ with real u, v. Since A and λ are real, equating parts gives $Au = \lambda u$ and $Av = \lambda v$. At least one of u, v is nonzero because $x \neq 0$; every nonzero one is therefore a real eigenvector.

Conclusion. The eigenvalue is real, and $\operatorname{Re} x$ is an eigenvector *if it is nonzero*. The printed unconditional claim about the real part needs this qualification: for $A = I_2$ and $x = (i, 0)^T$, the real part is zero and is not an eigenvector. If the real part vanishes, the nonzero imaginary part supplies a real eigenvector instead; multiplying x by $-i$ makes that vector its real part.

Exercise 5.5.25: Conjugation and real invariant directions

Write $x = u + iv$ with $u = \operatorname{Re} x$ and $v = \operatorname{Im} x$. A real matrix times a real vector is real, and linearity gives $Ax = A(u + iv) = Au + iAv$. Uniqueness of the real and imaginary parts of each component

therefore yields

$$\operatorname{Re}(Ax) = A \operatorname{Re} x, \qquad \operatorname{Im}(Ax) = A \operatorname{Im} x.$$

The real-entry hypothesis matters: a matrix containing i could mix the real and imaginary parts.

Exercise 5.5.26: Conjugation and real invariant directions

(a) Expand the eigenvector equation before separating parts.

Let $v = u + iw$, with u, w real, and use the given $\lambda = a - bi$:

$$A(u + iw) = (a - bi)(u + iw) = (au + bw) + i(aw - bu).$$

Since A is real, comparison gives $Au = au + bw$ and $Aw = -bu + aw$, the required identities.

(b) Match the two columns of the matrix products.

With $P = [u\ w]$ and $C = \left[\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right]$,

$$AP = [Au\ Aw] = [au + bw\ \ -bu + aw] = PC.$$

For completeness, u, w are independent when $b \neq 0$: if v were a complex scalar multiple of one real vector, the eigenvector equation would force that nonzero real vector to be mapped by the real matrix to a nonreal scalar multiple of itself, impossible. Thus P is invertible, which is the additional fact needed to pass from $AP = PC$ to $A = PCP^{-1}$.

Exercise 5.5.27: Two real invariant planes from complex eigenpairs

What we must show. Produce an exact real block factorization of

$$A = \begin{bmatrix} 26 & 33 & 23 & 20 \\ -6 & -8 & -1 & -13 \\ -14 & -19 & -16 & 3 \\ -20 & -20 & -20 & -14 \end{bmatrix}.$$

The characteristic polynomial factors as $(t^2 + 4t + 29)\,(t^2 + 8t + 116)$. An exact eigenspace computation gives the following eigenvectors for one root in each conjugate pair:

$$\lambda = -2 - 5i, \qquad v = \begin{bmatrix} -1 - i \\ i \\ 1 \\ 0 \end{bmatrix}.$$

$$\lambda = -4 - 10i, \qquad v = \begin{bmatrix} -1 + i \\ -2i \\ 2i \\ 2 \end{bmatrix}.$$

For each pair choose the root with negative imaginary part. Write its eigenvector as $u + iw$, then put u, w into two adjacent columns of P. The identities in Exercise 26 supply the corresponding 2×2

block of C. This gives

$$P = \begin{bmatrix} -1 & -1 & -1 & 1 \\ 0 & 1 & 0 & -2 \\ 1 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 \end{bmatrix}.$$

$$C = \begin{bmatrix} -2 & -5 & 0 & 0 \\ 5 & -2 & 0 & 0 \\ 0 & 0 & -4 & -10 \\ 0 & 0 & 10 & -4 \end{bmatrix}.$$

The exact determinant is $\det P = 2 \neq 0$, and multiplying the displayed matrices gives $AP = PC$ with zero residual in every entry. Therefore $A = PCP^{-1}$. The calculation can be reproduced with exact SymPy arithmetic: factor `A.charpoly().as_expr()`, then for each selected root `lam` compute `(A-lam*eye(4)).nullspace()`. Extract componentwise real and imaginary parts for the two columns. Multiplying those columns by a common nonzero real scalar does not alter the block identity, which is why fractional vectors may be cleared to integers.

Exercise 5.5.28: Two real invariant planes from complex eigenpairs

What we must show. Produce an exact real block factorization of

$$A = \begin{bmatrix} 7 & 11 & 20 & 17 \\ -20 & -40 & -86 & -74 \\ 0 & -5 & -10 & -10 \\ 10 & 28 & 60 & 53 \end{bmatrix}.$$

The characteristic polynomial factors as $(t^2 - 6t + 10)\,(t^2 - 4t + 29)$. An exact eigenspace computation gives the following eigenvectors for one root in each conjugate pair:

$$\lambda = 3 - i, \qquad v = \begin{bmatrix} -2 \\ 2i \\ -3 - i \\ 4 \end{bmatrix}.$$

$$\lambda = 2 - 5i, \qquad v = \begin{bmatrix} 1 - i \\ -4 \\ 0 \\ 2 \end{bmatrix}.$$

For each pair choose the root with negative imaginary part. Write its eigenvector as $u + iw$, then put u, w into two adjacent columns of P. The identities in Exercise 26 supply the corresponding 2×2 block of C. This gives

$$P = \begin{bmatrix} -2 & 0 & 1 & -1 \\ 0 & 2 & -4 & 0 \\ -3 & -1 & 0 & 0 \\ 4 & 0 & 2 & 0 \end{bmatrix}.$$

$$C = \begin{bmatrix} 3 & -1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 2 & -5 \\ 0 & 0 & 5 & 2 \end{bmatrix}.$$

The exact determinant is $\det P = -4 \neq 0$, and multiplying the displayed matrices gives $AP = PC$ with zero residual in every entry. Therefore $A = PCP^{-1}$. The calculation can be reproduced with exact SymPy arithmetic: factor `A.charpoly().as_expr()`, then for each selected root `lam` compute `(A-lam*eye(4)).nullspace()`. Extract componentwise real and imaginary parts for the two columns. Multiplying those columns by a common nonzero real scalar does not alter the block identity, which is why fractional vectors may be cleared to integers.